

mode revision note

October 2, 2026

finite sink for GB

In the previous model, the V/I concentration was set to be thermal-equilibrium concentration at the GB, which is a Dirichlet BC indicating the GB is a perfect sink for V/I:

$$C_V^{\text{GB}} = C_V^{\text{eq}}, \quad (1)$$

$$C_I^{\text{GB}} = C_I^{\text{eq}}, \quad (2)$$

With this Dirichlet BC, the results are mesh-dependent. With the refinement of the mesh, the defects profiles show a steeper gradient near the GB. Both the gradient and the magnitude of the solute segregation are higher than coarse mesh. These results are also observed in other RIS models using Onsager equation. [2] The width of RIS is also mesh-dependent, and do not converge with mesh refinement. The solution is to use a finite sink for the GB, which is a Robin BC:

$$J_V^{\text{GB}} = -k_V^{\text{GB}} (C_V^{\text{GB}} - C_V^{\text{eq}}), \quad (3)$$

$$J_I^{\text{GB}} = -k_I^{\text{GB}} (C_I^{\text{GB}} - C_I^{\text{eq}}), \quad (4)$$

where k_V^{GB} and k_I^{GB} are the sink strength for V/I at the GB. The sink strength and the defect-mediated transport coefficients are calibrated to reproduce both the experimental RIS magnitude and width.

References

- [1] P. Deng, Q. Peng, and E. H. Han. “Grain boundary oxidation of proton-irradiated nuclear grade stainless steel in simulated primary water of pressurized water reactor”. In: *Sci Rep* 11.1 (2021). Deng, Ping Peng, Qunjia Han, En-Hou eng England Sci Rep. 2021 Jan 14;11(1):1371. doi: 10.1038/s41598-020-80600-x., p. 1371. ISSN: 2045-2322 (Electronic) 2045-2322 (Linking). DOI: [10.1038/s41598-020-80600-x](https://doi.org/10.1038/s41598-020-80600-x). URL: <https://www.ncbi.nlm.nih.gov/pubmed/33446760>.
- [2] Russell Oplinger et al. “Radiation-induced segregation in dilute Fe–Cr: A rate-theory framework for the Cr enrichment–depletion transition at the grain boundary”. In: *Journal of Applied Physics* 139.15 (2026). ISSN: 0021-8979 1089-7550. DOI: [10.1063/5.0323296](https://doi.org/10.1063/5.0323296).

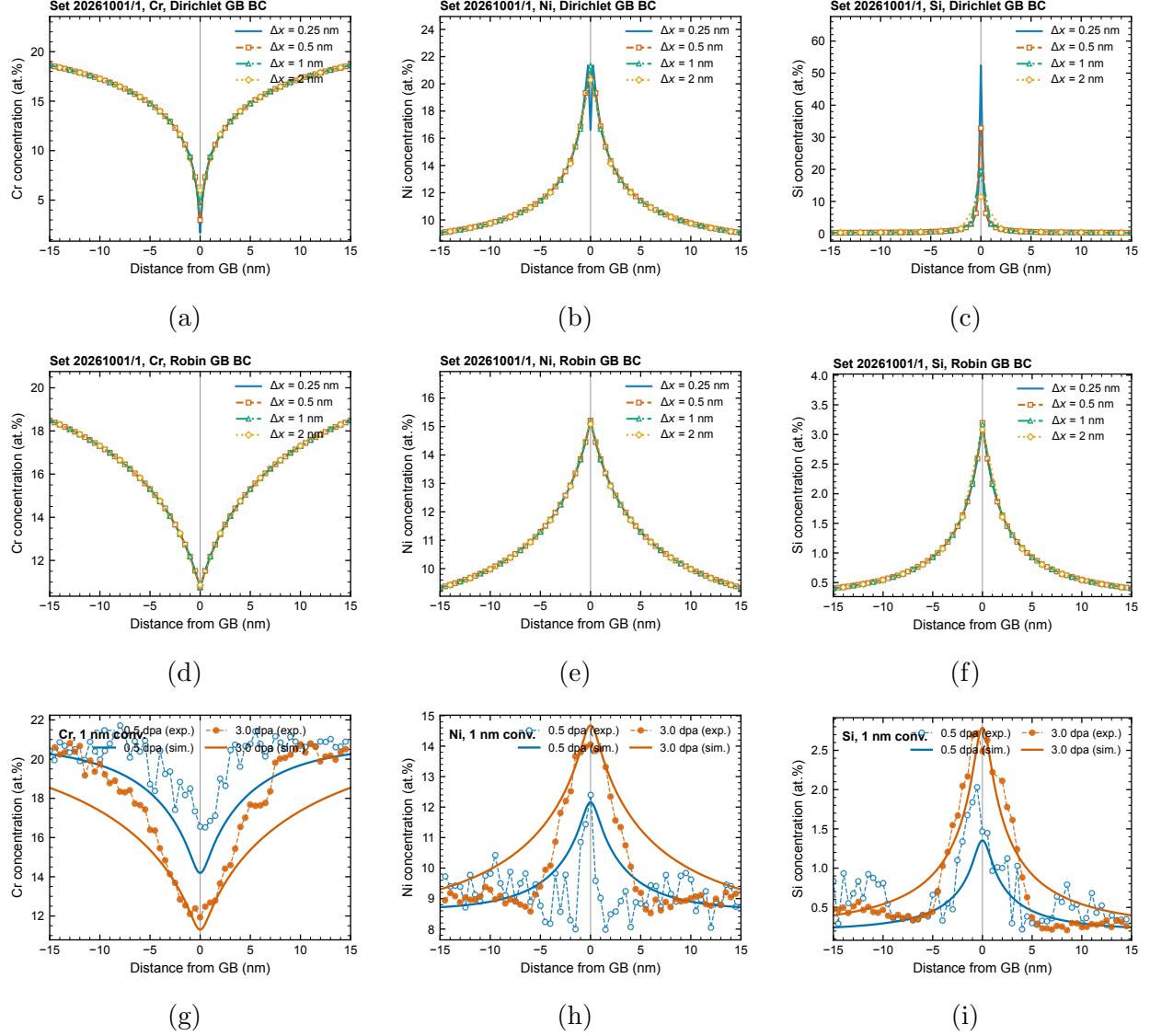


Figure 1: Comparison of RIS profiles with Dirichlet BC and Robin BC for GB. (a-c) show the RIS profiles with Dirichlet BC for different mesh sizes. (d-f) show the RIS profiles with Robin BC for different mesh sizes. (g-i) show the RIS profiles with Robin BC for different doses convolved with 1 nm FWHM compared with the experimental data[1].